

Puneet Dhaliwal

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Education

University of Toronto, St. George Campus – *BASc in Mechanical Engineering + PEY Co-op* 2023 - 2028 (expected)

- Entrance scholarship for academic excellence: \$5000; Dean's Honour List: Fall 2024
- Relevant Courses: Mechanical Design, Thermodynamics, Manufacturing, Fluid Dynamics, Materials Science

Experience

Aerodynamics Team Member, University of Toronto Supermileage – Toronto, ON Apr 2025 – Current

- Designed and developed the aerodynamic body of the U of T Supermileage Team's prototype vehicle in SolidWorks, creating assemblies and drawings used for manufacturing
- Performed CFD simulations in Ansys Fluent to guide design iterations, validating airflow behavior and improving aerodynamic performance
- Fabricated composite body panels through mold design, layup, and curing, translating CAD models into functional, lightweight structural components

CNC Machine Operator, EM Dynamics – Scarborough, ON May 2024 – Sept 2025

- Set up and operated CNC milling machines for aerospace and medical components to tolerances within ± 0.01 mm
- Interpreted detailed engineering drawings and CAD models, diagnosing machining deviations and performing root-cause analysis to adjust tooling, offsets, and machining parameters
- Supported safety and quality assurance processes, ensuring AS9100 compliance while maximizing machine uptime

Projects

3 Speed Go-Kart Jun 2024 – Aug 2024

- Engineered and fabricated a custom 3-speed go-kart, designing the chassis, steering, braking, and drivetrain systems through welding, machining, and 3D printing in collaboration with a partner
- Developed a 3-speed transmission in SolidWorks, researching gear design and torque ratios to optimize power delivery and vehicle performance
- Conducted structured speed, braking, and handling tests, achieving target speeds of 50 km/h while validating safety, durability, and performance standards

CNC Hobbyist Router Design Sept 2024 – Dec 2024

- Led a 4-person team over a 12-week timeline, coordinating weekly design reviews, task delegation, and scheduling (Gantt charts) to deliver a prototype-ready CNC router design
- Modeled the full CNC router assembly in SolidWorks, creating 60+ parts and 20+ detailed drawings covering joint interfaces, tolerances, and machining requirements
- Managed a \$4,200 budget to optimize parts and materials for manufacturability, creating a cost-effective assembly

Garage Entrance/Exit Jan 2024 – May 2024

- Led a 6-member team as Project Manager, using GANTT charts and weekly client updates to ensure all deliverables were met on time
- Developed and modeled a spiral ramp in SolidWorks, evaluating 120+ design alternatives and selecting a solution that reduced footprint by 67% compared to other concepts
- Tested using a 1:64 scaled 3D-printed prototype, validating continuous traffic flow and achieving the target 40-second descent time, presented to client

Skills

SolidWorks (CSWA Certificate), AutoCAD, Ansys Fluent, FEA, MatLab, C, Java, Python, Microsoft Office, 3D Printing, Project Management, Teamwork, Organization, Problem Solving, Communication